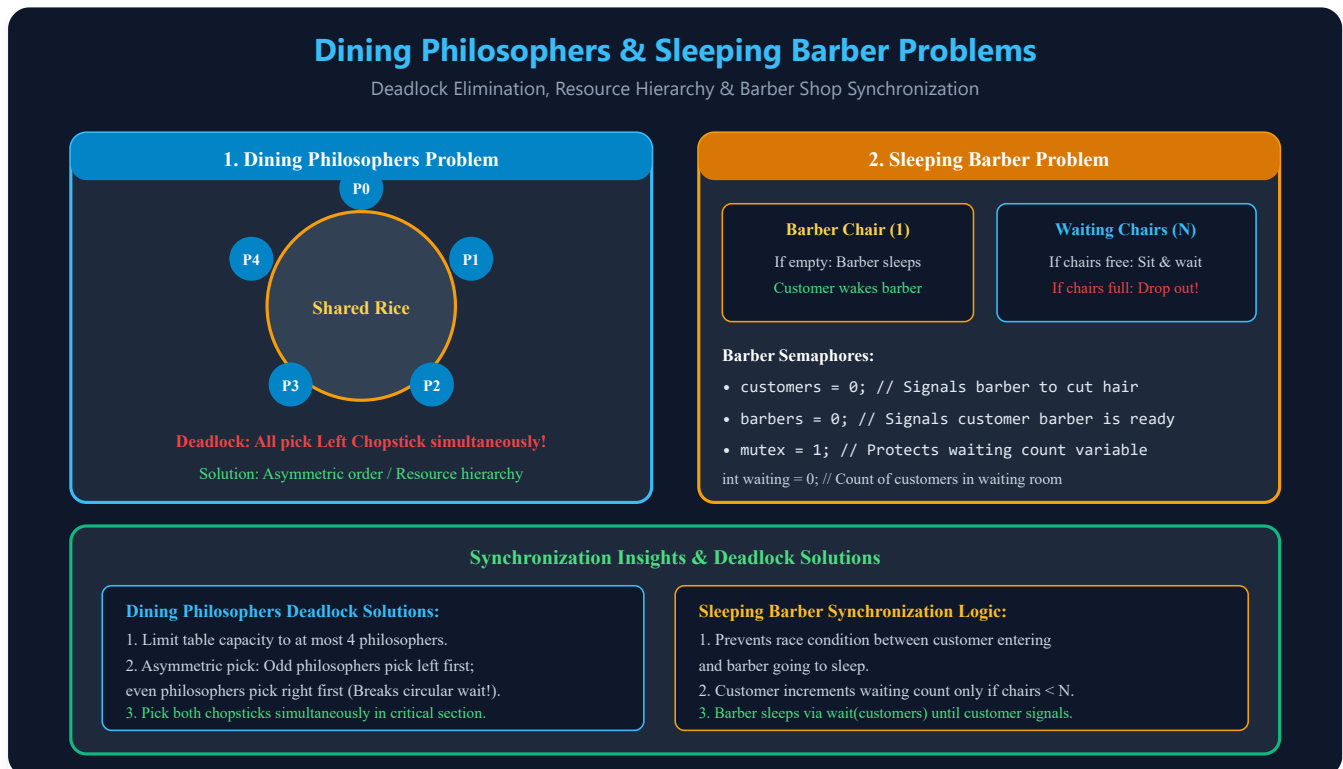


9. Classical Problems: Dining Philosophers & Sleeping Barber

Syllabus Coverage: Gateway Classes Slides 93–107 & AKTU Syllabus | Circular Wait Deadlock, Asymmetric Solution, Barber Synchronization.

9.1 Dining Philosophers Deadlock Prevention

Deadlock arises when all philosophers simultaneously grab their left chopstick. Solved using asymmetric pickup or limiting diners to [N-1](#).



9.2 Sleeping Barber Mechanics

- `customers = 0` : Barber blocks on `wait(customers)` until customer wakes him.
- `barbers = 0` : Customer blocks on `wait(barbers)` until barber chair is vacant.
- `mutex = 1` : Protects `waiting` counter from concurrent updates.